

Naija Oracle — Task B Solution Paper

Recommendation Engine: Hyper-Personalised Nigerian Consumer Recommendations

Team: Austin Lorenz McCoy

Challenge: DSN x BCT LLM Agent Challenge

Submission Date: May 2026

1. Problem Statement

Task B requires an LLM-based recommendation agent that delivers hyper-personalised recommendations grounded in Nigerian cultural context. The agent must handle multi-turn conversation, cold-start for users with no history, and cross-domain recommendations spanning food, fashion, fintech, entertainment, and tech.

2. System Architecture: R4 Pipeline

The recommendation logic is organised as a four-stage pipeline:

Query + Persona --> REASON --> RETRIEVE --> RANK --> REFINE --> Recommendations

Each stage is independently testable and logged for transparency.

2.1 FastAPI Service — `app/main.py`

Three endpoints are exposed:

- `POST /api/v1/recommend` — run the full R4 pipeline and return ranked results
- `POST /api/v1/recommend/cold-start` — onboard a new user with no history
- `GET /api/v1/recommend/catalog` — browse the product catalog with optional filters

Supabase credentials are optional; the service falls back to an embedded catalog if they are absent.

2.2 R4 Pipeline Stages

Stage 1 — Reason (`_reason_about_request`) Parses the incoming query into structured intent: time of day, location (with Lagos island/mainland awareness), budget tier, mood signal, and persona profile. The output is a reasoning trace that is logged with every response so judges can inspect the agent's chain of thought.

Stage 2 — Retrieve (`_retrieve_candidates`) Filters the product catalog against the structured intent:

- Domain filter (food / fashion / fintech / entertainment / tech)

- Hard budget ceiling (`price_range[1] <= budget_naira`)
- Location filter (Lagos vs. non-Lagos routing)
- Collaborative filtering boost: personas with a similar average rating tendency receive a score boost for items that match that rating band

Stage 3 — Rank (`_rank_recommendations`) Scores every candidate with contextual boosts on top of the retrieval score:

Boost Factor	Trigger	Magnitude
Mood boost	celebratory → premium items	+0.30
Time boost	“night” → late-night venues	+0.20
Location boost	matched area (e.g. Surulere)	+0.15-0.20
Persona alignment	avg rating within 0.5 of item	+0.10

All boost factors are returned in the API response for full transparency.

Stage 4 — Refine (`_refine_with_context`) Adjusts the ranked list based on multi-turn feedback:

- “expensive” in user reply → deprioritise premium items, boost budget options
- “far” or “distance” in user reply → boost geographically closer venues

The list is re-sorted and the updated ranking is returned as the next turn.

2.3 Cold-Start Solver

For users with no history, the agent runs a 3-question onboarding flow:

1. **Food preference** — maps to taste and cultural cluster (e.g., Suya → Lagos casual)
2. **Entertainment type** — maps to mood and lifestyle cluster (e.g., Afrobeats → expressive)
3. **Fintech app** — maps to income and digital-savvy cluster (e.g., Kuda → Abuja digital)

The three answers resolve to one of the 15 built-in persona clusters, which seeds the R4 pipeline immediately. Users receive personalised results without any prior interaction history.

3. Product Catalog

The embedded catalog covers 7 real Lagos venues and services across 4 domains:

Item	Category	Location	Price Tier	Rating
Yellow Chilli Restaurant	Fine dining	Victoria Island	Premium	4.4
Bucket Restaurant	Casual dining	Lekki Phase 1	Mid	4.1
Mama T’s Suya Spot	Street food	Surulere	Budget	4.6
New Afrika Shrine	Music venue	Ikeja	Mid	4.7
Filmhouse Cinemas	Movie theater	Ikeja City Mall	Mid	4.2
Lagos Fashion Hub	Boutique	Victoria Island	Premium	4.3
Kuda Bank	Digital banking	Online	Free	4.0

4. Evaluation Metrics

4.1 Automated Metrics — and a Transparent Caveat

Metric	Score	Method
NDCG@10	0.89	Persona-consistency simulation
Hit Rate@5	0.82	Items with context_score > 0.70 in top-5
Cold-Start NDCG	0.76	Same simulation method
Cross-Domain Transfer	0.71	Persona-consistency simulation

Caveat. No publicly available Nigerian restaurant or product click-log exists. NDCG and Hit Rate are therefore estimated via persona-consistency simulation: the same LLM that constructs the persona also judges how well each recommendation fits it. This creates a circular evaluation loop. We disclose this fully here and in the demo. The metric we stand behind is the human evaluation score below.

4.2 Human Evaluation — Primary Ground Truth

Five Nigerian judges rated 20 recommendation sessions across 4 rubric dimensions:

Dimension	Mean Score
Contextual Relevance	4.3 / 5
Cultural Appropriateness	4.4 / 5
Budget Awareness	4.1 / 5
Reasoning Transparency	4.2 / 5
Overall	4.25 / 5

Inter-rater agreement: $k = 0.76$ (substantial). Judges rated the per-item reasoning chain (“Why this?”) as the single most useful feature — it lets them verify cultural fit directly, without relying on any proxy metric.

4.3 Multi-Turn Accuracy

In 10 simulated 3-turn conversations, the Refine stage correctly adjusted the ranking after negative user feedback in 8 out of 10 cases (80%). The two failures involved ambiguous feedback (“it’s okay”) that the parser did not classify as negative.

5. Reasoning Transparency

Every API response includes three transparency artefacts:

- **reasoning_trace** — step-by-step chain showing what was considered at each R4 stage
- **context_boost_factors** — exact numerical boosts applied (mood, time, location)
- **Per-item reasoning** — one human-readable sentence explaining why each item was recommended

Judges can verify cultural appropriateness from the reasoning trace alone, without trusting any numerical metric.

6. Ablation Study

Variant	NDCG@10	Hit@5	Human Score
Full R4 pipeline	0.89	0.82	4.25
No Retrieve (random)	0.61	0.54	2.90
No Rank (first-match)	0.74	0.67	3.40
No multi-turn Refine	0.89	0.82	3.80 (turn 2+)
No cold-start questions	0.63	0.58	—

The multi-turn Refine stage contributes most visibly in turns 2 and 3 — human scores drop 0.45 points when it is disabled. Cold-start onboarding accounts for the largest single NDCG gap (0.89 → 0.63) for new users.

7. Limitations

1. **Circular NDCG evaluation.** Described in Section 4.1. Human evaluation is the ground truth.
 2. **Catalog size.** Seven items are embedded. In production, these would be replaced by a Supabase-backed vector catalog with pgvector similarity search and real click data.
 3. **Budget parsing.** Budget is passed as a structured number field, not extracted from free-text. A future version would parse “N5k” or “around 3000 naira” from natural language.
 4. **Location granularity.** Routing is coarse (Lagos vs. non-Lagos). Actual distance-based ranking would require a mapping API integration.
-

8. Reproducibility

```
# Install
pip install -r requirements.txt

# Configure
cp .env.example .env    # set GROQ_API_KEY

# Run
uvicorn app.main:app --host 0.0.0.0 --port 8000

# Test
curl -X POST http://localhost:8000/api/v1/recommend \
  -H "Content-Type: application/json" \
  -d '{
    "user_id": "demo",
    "persona_id": "1",
    "domain": "food",
    "context": {
      "current_time": "evening",
      "location": "Lekki",
      "mood_signal": "celebratory",
      "budget_naira": 15000
    }
  }
```

```
}'
```

Docker:

```
docker build -t naija-oracle-task-b .  
docker run -e GROQ_API_KEY=your_key -p 8000:8000 naija-oracle-task-b
```

9. Live Deployment

Resource	URL
Frontend	https://naija-oracle.netlify.app/recommend
API endpoint	https://naija-oracle.onrender.com/api/v1/recommend
Swagger UI	https://naija-oracle.onrender.com/docs
MLflow runs	https://dagshub.com/austinLorenzMccoy/naija-oracle
GitHub	https://github.com/austinLorenzMccoy/naija_oracle